Appx_15_On_Standard_Model_STUB_vX.X

Title: The Standard Model as a Survivable Realization
Status: Trailhead (STUB; mapping dictionary required before promotion)

Goal

Map the Standard Model (SM) into Proofield (PF) language without importing axioms by fiat—i.e., treat the SM as a survivable endpoint of ζ/ω/ξ-constrained geometry and AoC "constants-as-survivors," not as a starting assumption.

Core Claim

The SM is not "fundamental furniture," but the minimal stable interaction scaffold that survives the selection constraints encoded in PF. Gauge structure, particle content, and coupling hierarchy should be interpretable as emergent coherence regimes rather than arbitrary symmetry bookkeeping.

Must Include

- Sector dependency graph (not prose): QED ↔ electroweak ↔ QCD, with explicit interfaces (symmetry, charges, running, confinement).
- PF dictionary: translate SM primitives (fields, charges, gauge groups, symmetry breaking, mass terms) into PF concepts (constraint ζ, extension ω, chance ξ, survivorship classes).
- Units Firewall ledger: per sector, label quantities as [SURVIVOR], [EFFECTIVE], [TUNED], [CALIBRATED], or [UNRESOLVED] with AoC hooks.
- Anomalies / debts list as trailheads: mass gap, hierarchy/naturalness, confinement mechanism, CP violation, flavor/mixing, neutrino masses, baryogenesis, dark matter linkage, vacuum energy mismatch.
- Reduction checks: where SM must reduce from PF assumptions (no contradiction with known SM predictions).

Outputs

- SM-to-PF dictionary + translation table (one-page minimum, expandable).
- SM dependency graph usable by MoP trailheads.
- "What we still owe" list: which SM inputs are not yet Survivors in AoC/PF terms and what would promote them.

Primary Dependencies

- AoC_00_Atlas_Preface (status tags + Units Firewall discipline)
- AoC_3_On_137 (α) (and other AoC entries as they mature)

- Appx_11_On_Gauge_Structure_STUB
- Appx_17_On_Renormalization_and_Effective_Field_Theory_STUB
- Appx_13_On_Neutrinos_PF_v1.0 (neutrino sector mapping)

Primary Falsifier
The SM requires ad-hoc constants or structures that cannot be expressed as Survivors (or tightly bounded effective quantities) under AoC/PF discipline—i.e., PF cannot reproduce SM predictive content without smuggling in unexplained axioms.

Support
Appx_15_SM_Support/
(sector graphs, dictionary drafts, Units Firewall ledger, anomaly trailheads, mapping notes, rewrite targets, HoR packaging plan)

🤖 Start here if:
you are building the PF translation layer for particle physics, auditing which SM parameters remain “free knobs,” or constructing the anomaly trailheads that the AoC/selection program must eventually pay down.